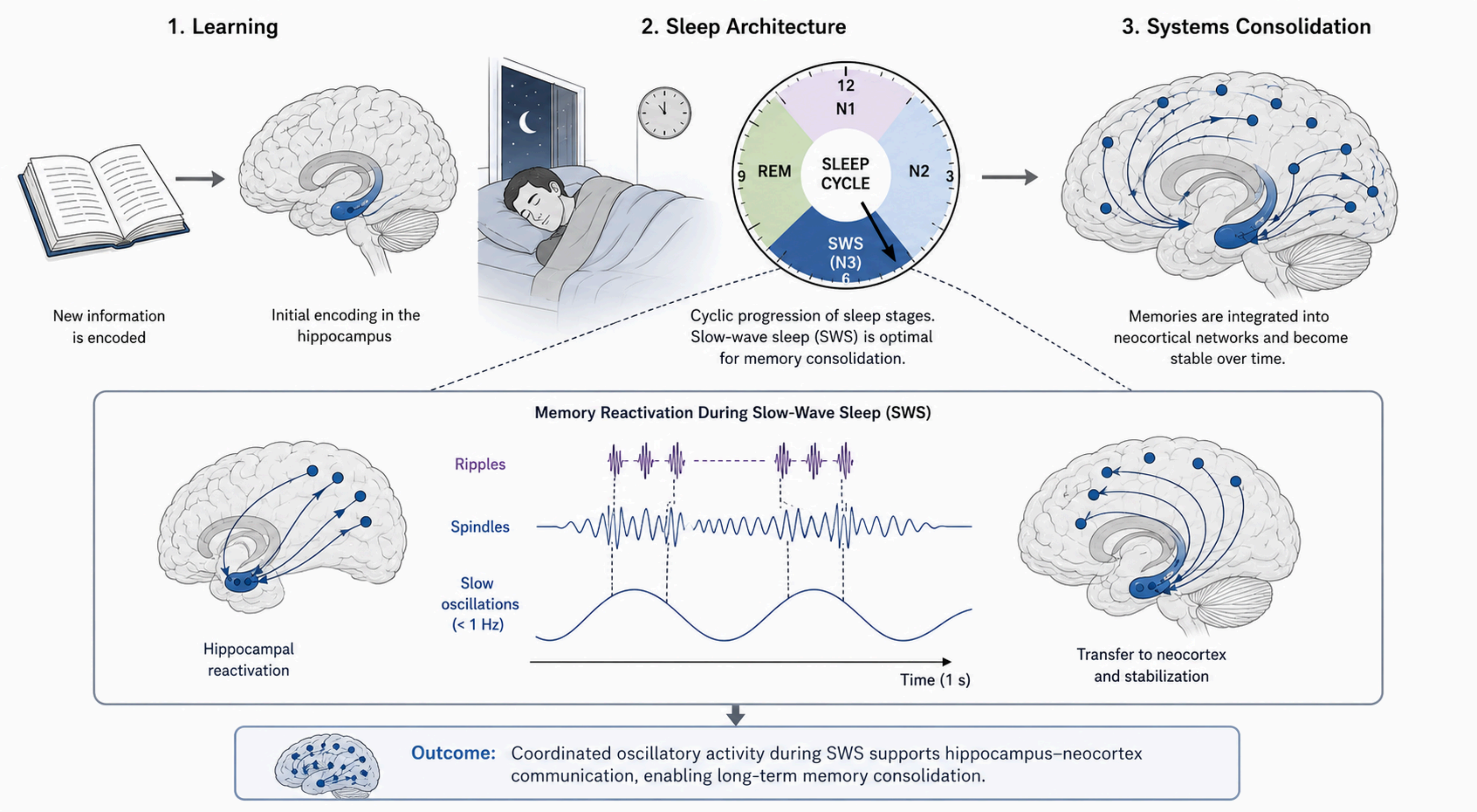

1. Learning
New information is encoded
Initial encoding in the hippocampus
2. Sleep Architecture
12
N1
N2
3
SWS (N3)
6
9
REM
SLEEP CYCLE
Cyclic progression of sleep stages. Slow-wave sleep (SWS) is optimal for memory consolidation.
3. Systems Consolidation
Memories are integrated into neocortical networks and become stable over time.
Memory Reactivation During Slow-Wave Sleep (SWS)
Ripples
Spindles
Slow oscillations (< 1 Hz)
Time (1 s)
Hippocampal reactivation
Transfer to neocortex and stabilization
Outcome: Coordinated oscillatory activity during SWS supports hippocampus–neocortex communication, enabling long-term memory consolidation.